Assessment of Seismic Damage of Buildings and Related Environmental Impacts

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Abstract: Sustainable development studies have recently begun to consider damage loss attributable to natural hazards. A sustainable perspective on building design entails an understanding of a structure's life-cycle environmental costs and consideration of the associated environmental impacts induced by natural hazards. Damage repair costs lead to additional material and energy consumption, creating harmful environmental impacts. Merging the results obtained from a natural hazard evaluation and related environmental impacts for buildings in seismically prone areas provides a novel outlook for sustainable design decisions. To evaluate the environmental impacts caused by buildings, the long-term effects accrued throughout a building's lifetime and the effects associated with damage repair need to be quantified. A literature review and a proposed method for this type of analysis have been developed and are discussed. Using *Athena* and *HAZUS-MH* software, this study evaluated the performance of steel and concrete buildings by considering their life-cycle assessments and earthquake resistance. This study determined that code design level greatly affects building repair and damage estimations. This study presented two case-study buildings and found specific results using several assumptions. Future research recommendations were provided to make this methodology more useful in real-world applications. Examining the cost and environmental impacts of a building using a cradle-to-grave analysis and seismic damage assessment assists in reducing material consumption and construction activities before and after an earthquake event occurs. **DOI: 10.1061/(ASCE)CF.1943-5509.0000584.** © *2014 American Society of Civil Engineers.*

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Introduction

"Sustainable development aims to enhance the quality of life by improving the social, economic and environmental conditions for the present and future generations," (Menna et al. 2013). Today, structural engineers are faced with the challenging task of balancing sustainable design practices with sufficient structural integrity for safety. Constructing and maintaining buildings consume vast amounts of energy and natural resources, and when natural hazard events (e.g., earthquakes) occur, repairing damage to buildings consumes additional energy.

Buildings are one of the largest consumers of natural resources and account for a significant portion of greenhouse gas emissions that contribute to climate change (DOE 1993). Fifty-four percent of energy consumption in the United States is related to buildings and their construction (Horvath 2004). This trend makes sustainable design a priority for building owners and designers. "In the United States, the building sector accounted for about 41% of primary energy consumption in 2010, 44% more than the transportation sector and 36% more than the industrial sector" (DOE 2013). A key aspect in sustainable development should be the building sector's energy consumption and material production, given the world's energy shortages and associated anthropogenic environmental effects. Few studies have linked the relationship between natural hazard mitigation and the environmental impacts caused by buildings. This study provides additional insights into the topic of environmental impacts inherent to buildings, including effects from earthquake-induced damages.

Life-cycle assessments (LCA) for buildings have become a major advancement in the design process for engineers. Numerous LCA studies have been completed for many different building types, and although such analyses are essential for sustainable design, they only consider a building's construction, use, and end-of-life phases. To incorporate new views on a building's energy consumption and contribute to future research, this paper explores the integration of the environmental impacts that buildings create, including damage repair from potential natural disasters (e.g., earthquakes). The motivation behind this research is largely driven by the worldwide depletion of natural resources and the increase in environmental concerns relating to natural hazard mitigation.

FEMA estimated that, on average, earthquakes cost the United States over $5 billion annually and, of that, California, Oregon, and Washington account for approximately $4.1 billion (77%) of this total estimated average annualized loss (Folger 2011). Several studies have examined structural building schemes and their resistance to earthquake events, although have not considered the environmental impacts incurred by repair and damage costs. Whereas both natural hazard mitigation and LCA evaluations have evolved during the past years, they are not closely related. Only limited research has linked the two concepts, although they both strongly relate to sustainable design.

Sustainable development is an issue recognized worldwide, and only recently has attention been directed toward the built infrastructure (i.e., buildings). Sustainable development has been a significant motivator and cause for the LCA methodology. LCAs provide key considerations for evaluating the environmental impacts produced by buildings throughout their life cycle. Low-impact

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buildings most commonly rely on advanced design techniques and innovative material technologies, both of which require additional upfront financial and environmental investments. Additional seismic resiliency increases initial construction costs but leads to lower repair damage (Comber et al. 2012). LCA studies considered all stages of a building's lifespan including initial construction, maintenance, and energy usage but rarely consider the effects caused by natural disaster events, which presents an opportunity for structural engineers to communicate the importance of a building's seismic risk and to employ mitigation strategies that minimize a building's life-cycle effect given these potential risks.

This paper provides a method for combining the results obtained from an LCA and earthquake damage assessment for buildings. Additionally, this study uses the proposed method to explore the differences between steel- and concrete-framed buildings built to various seismic code design levels, energy consumption, and greenhouse gas emissions when subjected to the same earthquake events.

The challenge of quantifying the environmental impacts from building and seismic damage is approached through two separate analyses. This study first compares the two types of structures using the LCA method and then uses HAZUS-MH (FEMA, Washington, DC) to determine the potential damage and repair costs under various seismic events. Next, this study explicitly examines the damage and repair costs caused by each building's structural components, which allows a clear comparison between the two building materials of steel and concrete. Although steel and concrete were compared, the method described throughout this paper may be used for all building types. This type of building examination is able to assists building owners and engineers in choosing between building designs and ultimately leads to environmentally conscious and seismically resilient design choices.

Background

A study completed by Menna et al. (2013) presented a novel approach of including seismic damage into their building LCA. Their study concluded that the evaluated seismic events contributed 6% to the building's overall life-cycle effect and 25% was accounted for by seismic damage when compared with the building's initial construction phase. Taghavi and Miranda determined that, in a typical building, the structural system accounts for approximately 10–20% of the construction cost (Menna et al. 2013; Taghavi and Miranda 2003). Although the structural system is a small portion of the total building cost, upgrades to the system could result in cost savings after a seismic event.

The Structural Engineers Association of California (SEAOC) represents the structural engineering community and states in their Blue Book that seismic design is, "primarily to safeguard against major structural failure and loss of life, not to limit damage or maintain function" (Kneer and Maclise 2008; SEAOC Blue Book 1999). Therefore, a building designed using code minimums does not necessarily ensure consideration of the entire building's life-cycle effects and the consequences of damage and repair (Kneer and Maclise 2008). Kneer and Maclise (2008) summarized case studies that examined the role of building performance in minimizing the environmental impacts for buildings located in areas of seismic risk and summarized available software tools. Kneer and Maclise (2008) suggested that additional case studies are needed to expand and increase the robustness of research concerning LCAs. Software such as *HAZUS-MH* (FEMA 2002), a performance-based design tool, and *Athena* (Athena Sustainable Materials Institute, Ontario, Canada) a LCA tool, were recommended.

A study written by Comber et al. (2012), examined the need to "shift away from designing code-minimum buildings that are life-safe but often disposable," and presented a method that examines a building's long-term environmental impacts, including expected seismic damage. The authors completed a comparative study that examined two case-study buildings—a concrete moment frame and a shear wall system—and determined that the moment frame had lower total carbon equivalent emissions. The study concluded that the moment frame required 19% of the building's total embodied energy to repair damages, and the shear wall system required 15%. The study determined that although the shear wall system had more associated carbon outputs considering its initial embodied carbon output, the moment frame had less carbon output in seismic repair damages (Comber et al. 2012). As shown through Comber and Sinclair's study, a project that has a greater initial investment and increased carbon outputs attributable to structural choices ultimately reduces the carbon outputs caused by repair damages after an earthquake event (Comber et al. 2012). The study concluded by recommending that the structural community explore this notion of determining the environmental implications of performance-based and disaster-resilient design (Comber et al. 2012).

Analyses on building materials and their environmental impact is essential for comparing various design options, improving existing practices, and recognizing opportunities for environmental improvements. To date, LCA provides the most complete framework for aiding in the decision-making processes needed to accomplish substantial environmental improvements. Life-cycle assessment, also known as life-cycle analysis or cradle-to-grave analysis, is a tool used to assess a building's environmental impact. A LCA takes a broad overview of a building's lifespan and assesses the potential environmental impacts during different life-cycle stages.

History and Methodology

According to the ISO standards, the LCA method is "a technique for assessing the environmental aspects and potential impacts associated with a product" (ISO 14040 1997). A complete LCA of a building includes material extraction, processing and manufacturing, construction, use, end-of-life, and transportation effects related to all of these stages. A broad set of environmental impacts is found using the LCA analysis method, such as global warming potential, resource depletion, toxicity, and ozone depletion. A complete LCA consists of four phases that are defined by the International Standard for Environmental Management (ISO 14040 1997). The goal and scope phase defines motives for completing the LCA and details the results for specific audiences. The inventory phase includes the collection of all data needed for environmental calculations. Examples of LCA inputs may include energy, water, land use, and natural resources. Outputs may include harmful environmental emissions into the air, land, or water, and use of nonrenewable resources. The main intention of the impact assessment phase, which assigns a numerical value to the environmental impacts, is to consider the relative magnitude and significance of each environmental impact considered on the basis of the inputted and outputted material data collection. For example, the carbon emissions most commonly reported from a LCA are given in CO_2 equivalents. All inputs and outputs that contribute to a building's overall greenhouse gas emissions are multiplied by a CO_2 equivalence or characterization factor. The conversion allows the user to compare different inputs and outputs with a common greenhouse gas unit. The interpretation phase and the last phase in a complete LCA interprets the impact results. The user identifies significant contributors to a building's overall environmental impact, draws

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conclusions, and determines associated limitations to assist in future recommendations for environmental considerations.

LCAs group a building's lifespan into separate phases: initial construction, usage, and end of life. The initial construction phase, also known as a building's embodied energy, includes material extraction, manufacturing, and assembly, and the transportation effects related to each. The use phase, also known as the operational energy usage phase, requires the analyst to estimate a building's anticipated energy use over an assigned period. The last life-cycle stage considered in a complete LCA of a building includes the end-of-life phase. The data related to demolition and solid waste activities are required. Each stage needs to be considered when completing a full LCA for a structure.

LCA has become common and is increasingly being used as a design aid. LCAs are capable of providing decision makers with the ability to weigh the environmental benefits of one design with that of another. An ever-popular use for LCA is in union with the Leadership in Energy and Environmental Design (LEED). The U.S. green building council defines LEED as "a voluntary, consensus-based, market-driven program that provides third-party verification of green buildings." An increasing number of owners and designers are becoming aware of LEED certification. LEED focuses on reducing a building's environmental impact considering its initial construction, and on reducing a building's operational energy usages (LEED 2012).

Past Life-Cycle Studies for Buildings

Guggemos and Horvath (2005) completed a comparative study between a concrete-frame structure and a steel-frame structure by isolating each of the LCA stages. Both structural frames were designed to model typical office buildings located in the Midwest United States with a projected lifespan of 50 years. The buildings shared interior finishes, painted partition walls, built-up roofing, and concrete mat foundations, and only their structural frame varied. Guggemos and Horvath (2005) determined that the steel-frame structure was more costly than the concrete-frame structure using (R.S. Means 1999) for their cost estimates. The authors compared the two structural frames at the two levels of initial construction phase and overall LCA. The study determined that during the construction phase, the concrete structural frame had greater associated energy use and CO_2 emissions, primarily because concrete uses more temporary materials, has a longer installation time, and produces more transportation effects than does steel. Additionally, the study determined that the overall life-cycle comparison of both buildings seemed to have very similar environmental impacts (Guggemos and Horvath 2005).

A building's energy consumption is divided into embodied energy and operational energy. Embodied energy is the amount of energy used to construct, maintain, and dispose of a structure. Embodied energy includes the effects from obtaining the raw materials, the processes that these materials go through, the assembly of these materials, the maintenance and operations required to maintain those materials, the effects of disposing the product after its useable lifespan, and transportation effects related to each. A building's operational energy includes heating, water, air-conditioning, and lighting. Cole and Kernan (1996) determined that approximately 80–90% of a building's energy usage is accredited to operational energy alone. The study recognized that, "An important conclusion is that published studies on initial embodied energy of buildings provide a guide to the typical ranges for the initial embodied energy of office buildings, however it is difficult to interpret and compare studies in any detail because of the lack of definition of what was included within the total embodied energy figures" (Cole and Kernan 1996). The study also concluded that the operational energy usage in buildings represents the largest component of life-cycle energy usage. "As environmental issues continue to become increasingly significant building design priorities, we can anticipate considerably improved energy standards" (Cole and Kernan 1996). Additionally, as the operational energy is decreased, more attention is drawn to reducing a building's embodied energy. The study also suggested that reducing a building's embodied energy involves much more than comprehensive design approaches; it also involves examination of a building's repair and replacement costs. The study ended by suggesting that future research focus on material longevity and the ability to replace elements within a total building assembly.

Earthquake Building-Induced Damage

Earthquakes are devastating events and incur billions of dollars in repair costs. Engineers' understanding of earthquakes has improved significantly during the past twenty years. This better understanding is primarily the result of a significant improvement in the technology used to track and collect data on seismic events. All data are collected and shared globally, giving scientists and researchers the opportunity to develop more complete models of the earth's ground motion attributable to seismic events and to use this information to design safer buildings. The USGS stated that, "In 1931, there were about 350 stations operating in the world; today, there are more than 8,000 stations and the data now comes in rapidly from these stations by electronic mail, internet and satellite." Major earthquake events such as San Fernando 1971, Loma Prieta 1989, and Northridge 1994 have provided engineers with excellent seismic data for analyzing future earthquake occurrences and understanding structural responses.

Estimating the potential damage that a building experiences under a given earthquake event is challenging. Several studies have turned to computer software programs to help estimate seismic damage to buildings. *HAZUS-MH*, a nationally accepted computer software program that estimates potential losses from earthquakes, floods, and hurricanes, eases this challenging analysis (FEMA 2002). FEMA developed *HAZUS-MH* with the National Institute of Building Sciences (NIBS) to aid with hazard mitigation. The program uses up-to-date and current engineering and scientific knowledge to assess loss estimates for earthquake hazard events. *HAZUS-MH* has default databases that include data regarding building inventory and demographics for all regions of the United States (Kircher et al. 2006). The program provides historical data on past seismic events that have occurred in the United States and, additionally, allows for the creation of a theoretical event. When a user inputs a certain earthquake event, the potential earth science hazard (PESH) module estimates ground motion and ground failure on the basis of the fault type and location, and the earthquake magnitude—each of which may be selected by the user. After an earthquake scenario is created, a direct damage module is created that estimates the damage in terms of the probability of exceeding states of set damage for any given ground motion or failure.

Methodology

The flow diagram shown in Fig. 1 describes the methodology of this study. As shown, the software used in this study, *Athena* and *HAZUS-MH*, produces results that are then combined to provide the total environmental impact.

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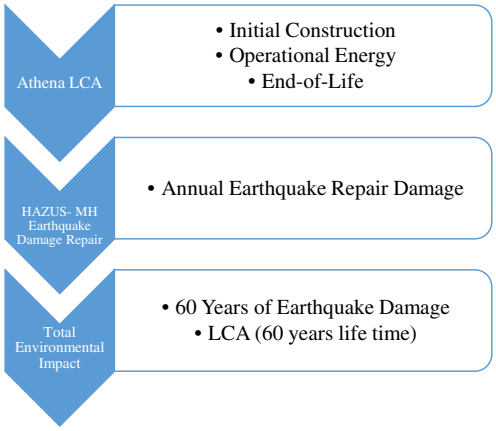


Fig. 1. Flow diagram combining earthquake damage with life-cycle analysis

Athena Life-Cycle Analysis

The writers considered a number of LCA software tools, and selected the *Athena* eco-calculator (AEC), explicitly because material and quantity-takeoff data were not needed for each case study building. *Athena* LCA software has two LCA calculators available: the *Athena* eco-calculator and the *Athena* impact estimator (AIE) (*Athena*). AEC permits a user to complete an entire building's LCA by allowing the user to input building type assemblies and location. The program has built-in material databases for each stage of a building's life cycle, including raw materials, material manufacturing, transportation, on-site construction, maintenance, repair and replacement, demolition, and disposal. The AEC is similar to the AIE but does not include operational energy usage (*Athena*). A complete LCA for each case study building is accomplished by using the AEC for the environmental impacts caused by initial construction and end-of-life impacts, and by using the AIE to determine each building's operational energy effects.

The *Athena* eco-calculator requires square footages for each assembly included within the LCA, such as foundation and footings, columns and beams, intermediate floors, exterior walls, windows, interior walls, and roof. The AEC outputs environmental impacts, including fossil fuel consumption, global warming potential, acidification potential, human health respiratory effects potential, eutrophication potential, ozone depletion potential, smog potential, and weighted resource use. AEC impact categories are set forth by the EPA for LCA and are in accordance with ISO 14040. This paper examines fossil fuel consumption and global warming potential differences between the two case study buildings.

The AIE was used to determine each building's operational energy usage. The operational energy for both buildings was determined assuming that each building consumes the same amount of energy per year. Cole and Kernan (1996) assumed that, "The difference in the operating energy between wood, steel and concrete framed buildings is negligible." This assumption is adopted in this paper because two case study buildings analyzed have the same internal building materials and only their structural frame varies. The AIE allows the user to input the location and the estimated annual fuel consumption quantities per square floor area, such as for natural gas and electricity. After providing the building's location and consumption information, the software generates the appropriate electricity grids, transportation modes and distances, and product manufacturing technologies that affect the building's environmental impact. The annual assumptions for energy consumption for both case study office buildings located in the United States were 183 kWh/m^2 and 10 m^3 of natural gas per square meters (DOE 1983). The annual consumption data were entered into the *Athena* Impact Estimator and the operational energy effects were obtained.

Earthquake-Induced Damage

"HAZUS-MH, Hazards United States Multi-Hazard, is a comprehensive software tool developed by the Federal Emergency Management Agency (FEMA) of the United States through the National Institute of Building Sciences (NIBS), to determine multi-hazard loss estimations in the United States on a regional basis" (Ploeger et al. 2010). HAZUS earthquake has two building damage functions: capacity curves and fragility curves. Each function is given for low-, medium-, or high-rise buildings of varying types (e.g., steel, concrete). The capacity curves utilize a structure's yield limit and ultimate strength, and characterize nonlinear (push-over) structural behavior. For each building type, the capacity parameters change and illustrate the different levels of seismic design and anticipated seismic performance. Fragility curves describe the probability of damage to a structural system and its nonstructural components (sensitive to drift or acceleration). Fragility curves are a good indicator of how a structure withstands a seismic event and provide insight into damage and repair costs attributable to certain ground motion events. For a given building type and its response, fragility curves distinguish damage in four main categories: slight, moderate, extensive, or complete.

Three seismic design code levels—high, moderate, low—and Pre-Code form the basis for the damage functions within HAZUS. The Pre-Code design level represents structures built before 1941, when seismic codes were not required for buildings in seismically active regions. High, moderate, and low code design levels utilize the 1994 uniform building code and lateral force design requirements of seismic zones 4, 2B, and 1, respectively (Kneer and Maclise 2008). Buildings built after 1975 are considered high seismically designed and buildings built between 1941 and 1975 are considered moderate seismically designed.

The seismic events selected for this study are Northridge, a 100-year event, and an annualized damage analysis for a building in Los Angeles, California. On January 17, 1994, a 6.7 magnitude earthquake hit the San Fernando Valley. The Northridge earthquake was produced by a strike-slip fault having a subsurface and surface length of 36 and 24 km, respectively. The earthquake had an orientation of 0° and a dip angle of 90 km. The epicenter of the Northridge earthquake was at a latitude and a longitude of 34.21 and −118.537. The depth and width of the earthquake were 2 and 10 km, respectively. The Northridge earthquake caused 60 deaths and 7,000 injuries, and left 20,000 people homeless in Los Angeles, Ventura, Orange, and San Bernardino Counties (USGS 2013). The total cost of the damages was estimated at between $13 and $20 billion (USGS 2013). Because of the severity and location of this damage, this event was chosen to model the greatest damage that could occur to a building from a single earthquake event.

The 100-year return period provides an estimate of the likelihood of a certain magnitude event occurring. A 100-year event has a 0.01% probability of occurring in any given year during a building's lifetime. *HAZUS-MH* uses a default value of a magnitude 5 earthquake for the 100-year event scenario. *HAZUS-MH* does not allow the user to choose a 60-year event that corresponds to each building's lifespan. The 100-year event was chosen to best represent a likely event that could happen to each type of structure.

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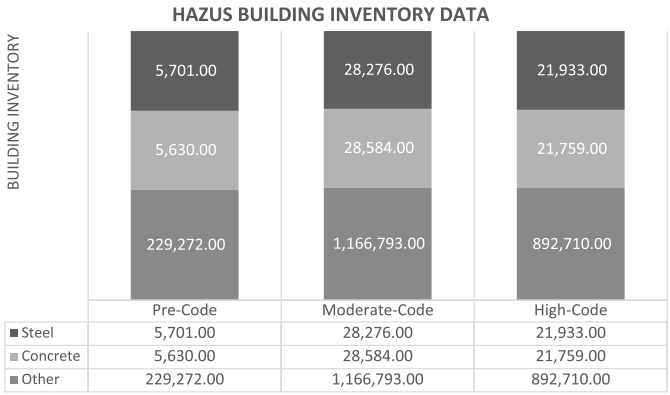


Fig. 2. *HAZUS-MH* building inventory (*HAZUS-MH*)

The annualized earthquake loss estimation determined using *HAZUS-MH* represents the estimated long-term value of earthquake damage and losses to the general building stock in any single year in a specified geographic area (FEMA 2008). *HAZUS-MH* takes into account such factors as historic patterns of frequent smaller earthquakes with infrequent but larger events, and provides a balanced estimate of potential earthquake damage. This feature allows a comparison of buildings built to different codes that are located in different regions around the country (FEMA 2008).

The two buildings in this study are in Los Angeles County, which consists of approximately 4,083 square miles and includes 1,652 census tracts (Kircher et al. 2006). Fig. 2 shows the default inventory data included in *HAZUS-MH* for the Los Angeles County region. For each design level and structure type, *HAZUS-MH* provides the default inventory building count (FEMA 2013). The total building count shown in Fig. 2 includes all building types; i.e., wood, steel, cast-in-place concrete, precast concrete, reinforced masonry, unreinforced masonry, and manufactured homes. As shown, steel and concrete buildings make up a small portion of the overall building inventory for this area.

HAZUS determines the probability of damage to the general building inventory and converts these probabilities into the number of damaged buildings under the four damage states. This study uses these probabilities of damage for the different building types in the general region to describe the damage that would occur to an individual building.

The total probability of damage to the general population of steel and concrete buildings was determined for each of the previously noted earthquake events. The probability of damage was provided for each of the four damage states, each building type, and each design code. The *HAZUS-MH* technical manual provides structural repair cost ratios in percent of building replacement costs for each damage state and building type. This analysis assumed that both building types under consideration were COM4 in the *HAZUS-MH* technical manual. COM4 is a group of buildings consisting of generic professional business offices (FEMA 2013). Although HAZUS provides a cost estimate for COM4 building types, a more up-to-date cost estimate for each steel- and concrete-framed structure was used. REED Construction Data Inc., an online source, provided a cost estimate for each building type. Each structure was analyzed as a four-story (each story being 10 ft) office building in Los Angeles, California with total square floor area of 86,400. The REED building cost estimates are derived from a building model built in 2013 that assumed basic components and did not include a basement. The steel building's approximate cost was estimated to be \$225/sq ft, and the approximate cost for the concrete building type was \$184/sq ft (Reed Construction Data 2013). Both cost estimates were used when calculating the damage cost related to the percent damage on the basis of the different probabilities and their associated damage states.

Table 1. Repair Cost Ratios for Each Damage State

Structural damage state	Structural repair cost ratios (in percentage of building's replacement cost)
Slight	0.4
Moderate	1.9
Extensive	9.6
Complete	19.2

For building type COM4, Table 1 presents the associated structural damage state and the associated structural repair cost ratios (FEMA 2013) for each damage state. The structural repair cost ratio relates building damage to total building cost.

The structural repair cost ratios and buildings' total costs were used to determine the cost of the damage under each different damage state.

Incorporating Seismic Damage in a Life-Cycle Assessment

To calculate the environmental impacts caused by seismic damage, this study related the building's embodied energy to its cost and used this relation to determine the cost of damage and environmental impact for repairs. *HAZUS-MH* provided the damage probabilities for each damage state, which were then multiplied by each of the structural damage ratios and summed for all separate damage states to obtain each building's total probability of damage. The sum of the damage probability was then multiplied by the total cost of the building to obtain the total cost of damage. To relate the cost to energy, the AEC result for each building was divided by its total cost to obtain the energy in MJ per dollar. This relationship was used to determine approximately how much energy each event consumes to repair the damage. The flow diagram in Fig. 3 summarizes this process.

The two single events analyzed in *HAZUS-MH* (i.e., Northridge and a 100-year event) contribute to a building's overall LCA effect. This study examined the *HAZUS-MH* annualized loss to obtain a per-year potential damage estimate for each building type. The annualized damage was multiplied by the estimated lifespan of each building to represent 60 years of seismic damage that each structure could experience. The annual seismic damage cost was also converted into MJ of energy and summed with the overall life-cycle effect of each building.

Case Study

Life-Cycle Analysis Results

The two case study buildings under consideration were modeled as low-rise commercial buildings in Los Angeles, California. Both office buildings were analyzed as four stories, with each story height being 10 ft. Each story is 120 by 180 ft, for a total floor area of 86,400 sq ft. Both office buildings were assumed to share the same foundations, interior walls, and window material; thus, these items were not included in the environmental comparison. The structural elements in a building have a stronger environmental effect because they have the higher percentage of material quantity when compared with the entire building envelope (Delong et al. 2011).

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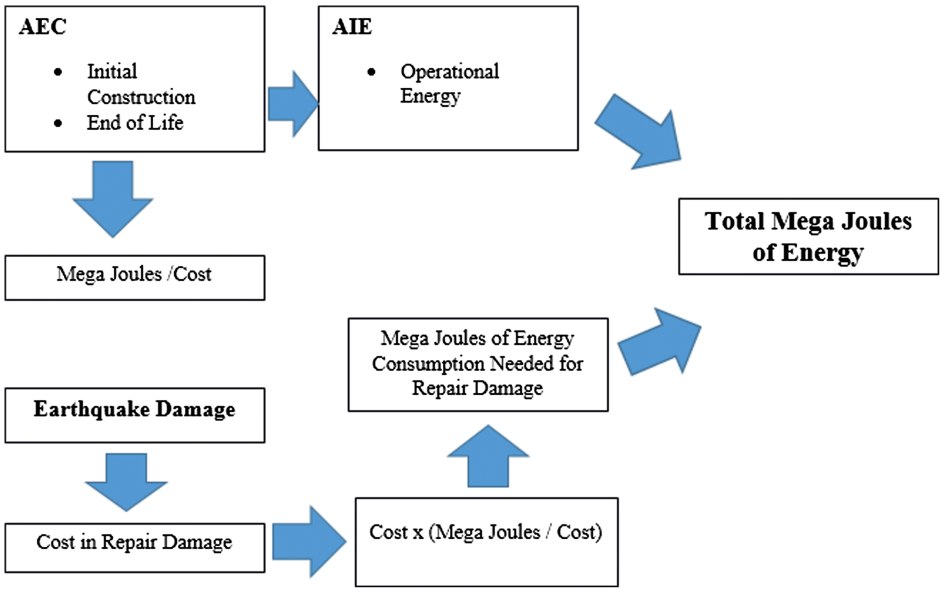


Fig. 3. Flow diagram of method for calculating total environmental effect

Table 2 provides each assembly type and area used by the *Athena* Eco-Calculator program.

Figs. 4 and 5 provide the environmental impact results for both the concrete and the steel generic building types.

Table 3 provides the energy consumption and global warming potential per square feet for comparison purposes.

The differences between the energy consumption of the generic steel and concrete building, not including each building's operational energy, illustrates that steel has less of an effect. The steel building was determined to have a lower fossil fuel consumption and global warming potential compared with the concrete building. However, this result could be attributable to the material assembly choices made within AEC. The results from AEC were obtained using several assumptions that should be considered in future studies. A more rigorous analysis of detailed building assembly components should be conducted if this method is to be used in a real-world design project.

Table 4 shows the operational energy results obtained from AIE. The results show that location affects the calculated operational energy usages. The methods used to extract, refine, and distribute energy to buildings vary throughout America. The Guggemos and Horvath (2005) operational energy was shown to compare them with the AIE estimate energy consumption. The results are similar because both this study and the Guggemos and Horvath (2005) study had similar usage inputs.

Table 5 provides each structure's operational energy and embodied energies.

Approximately 98% of the total energy consumption was solely the result of the operational energy when considering the overall LCA effects for both building types. These data are comparable to the findings of Cole and Kernan (1996). Although operational energy is the main issue in a building's energy consumption, the intent of this research was to examine each building frame and its resiliency and effect when subject to seismic events. The implications and recommendations for future research are subsequently discussed in the results discussion section of this paper.

HAZUS-MH Results

Steel Building HAZUS-MH Results

Table 6 provides the information used in the damage analysis for the steel building.

Table 7 describes the damage to the general population of steel buildings if a Northridge event occurs in the Los Angeles, California area. For a general population consisting of high designed steel buildings, *HAZUS-MH* estimates that 10.41% of the population will experience slight damage, 4.66% will experience moderate damage, 0.63% will experience extensive, and 0.03% will be completely destroyed. On the basis of these results, buildings that are built to higher standard code regulations exhibit drastically reduced associated repair and damage costs. The additional energy needed for repair would contribute to an individual steel building's

Table 2. Case Study Building—Athena Input Data

Assembly	Steel building	Concrete building	Input	Units
Columns and beams	HSS column/WF beam	Concrete column/concrete beam	86,400	SF
Intermediate floors	Steel joist	Suspended concrete slab	64,800	SF
Exterior walls	Steel cladding (26 ga) R—3.8 continuous insulation sheathing, 2x4 steel stud 24" o.c	Precast concrete cladding—cast in place concrete continuous insulation + latex paint	19,080	SF
Roof	Steel roofing system R- 20 continuous insulation + polyethylene membrane, open-web steel joist w/steel decking, gypsum board + latex paint	Suspended concrete slab-EPDM membrane R- 20 continuous insulation + polyethylene membrane + latex paint	21,600	SF

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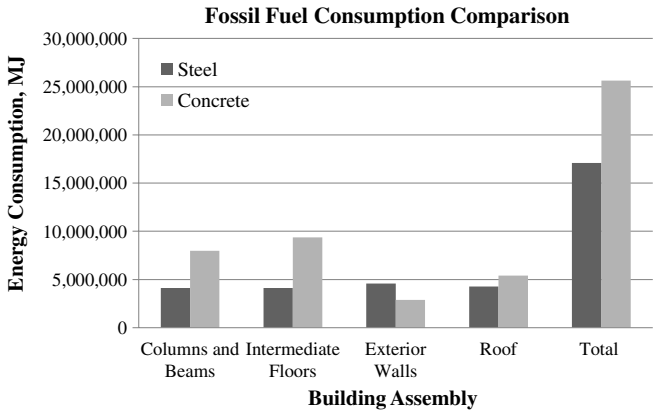


Fig. 4. Fossil fuel consumption comparison (data from *Athena*)

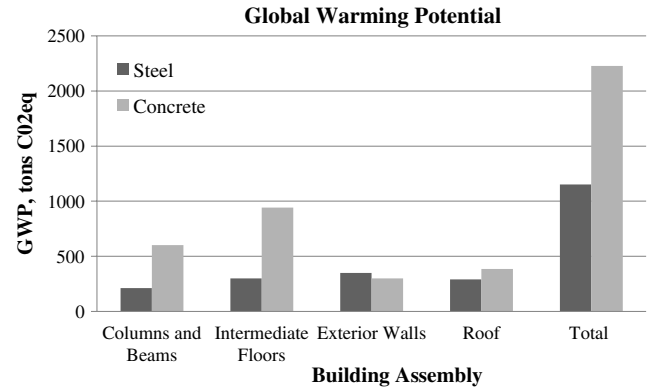


Fig. 5. Global warming potential (data from *Athena*)

life-cycle impact and should be considered in the design phase of a project. The estimated energy consumption for each building design code is an underestimate of its total impact because each LCA included only each building's structural frame.

Table 8 shows that 23.34% of the region's steel buildings, built to high code standards, will experience slight damage, 11.39% will experience moderate damage, 1.19% will experience extensive damage, and 0.03% will experience complete destruction under a 100-year return event, resulting in total costs to repair the damage of $83,640 and energy consumption of 73,440 MJ. As the building design code standard decreases, these numbers increase.

Table 9 provides the average annual damage probabilities that could occur to the general steel building population. In one year, a typical steel building will experience $4,673 of damage repair costs and will consume 4,103 MJ of energy.

Table 3. Athena Eco-Calculator LCA Results

Building type	Fossil fuel consumption (MJ/SF)	GWP (ton CO_2eq/SF)
Steel	197.78	0.013
Concrete	296.73	0.026

Note: Data from *Athena*.

Table 4. Operational Energy Comparison

Study	Energy (MJ/SF)	Description of location
Athena impact estimator	9,214	U.S.
Athena impact estimator	8,751	Los Angeles, CA
Guggemos and Horvath (2005)	6,862	Midwest U.S.

Table 5. Overall LCA Results Comparison (Data from Athena 2013)

Building type	Embodied energy (MJ/SF)	Operational energy (MJ/SF)	Total LCA environmental impact (MJ/SF)
Steel	198	8,751	8,949
Concrete	297	8,751	9,048

Note: Data from *Athena*.

Table 6. *HAZUS-MH* Steel Data for Damage Cost Calculations

Damage relationship	Value
Energy per replacement cost (MJ/$)	0.88
Total cost replacement value ($/SF)	225.25
Energy for building, 60-year lifespan (MJ/SF)	197.78

Concrete Building HAZUS-MH Results

Table 10 provides the information used in the damage analysis for the concrete building.

The energy per replacement cost for the concrete building is approximately 50% more than that of the steel building. This result is because of the steel building's higher cost and lower calculated environmental impact; however, this study made several assumptions, indicating that these results are specific to this case.

Table 11 describes the damage that would occur to the general concrete building population if a Northridge event hit the Los Angeles, California area.

Table 12 provides the data on the damage that would occur to the general concrete building population within Los Angeles, California if a 100-year return event occurred.

Table 13 shows the annualized seismic damage that a concrete building could experience in any given year.

Table 7. Steel Building Damage Cost from Northridge Earthquake

Design code	Damage state	Probability of damage (precentage)	Damage ratio (precentage)	Damage cost ($)	Repair energy (MJ)
High	Slight	10.41	0.0004164	38,226	33,565
	Moderate	4.66	0.0008854		
	Extensive	0.63	0.0006048		
	Complete	0.03	0.0000576		
	Total		0.0019642		
Moderate	Slight	9.09	0.0003636	76,385	67,069
	Moderate	6.87	0.0013053		
	Extensive	1.85	0.001776		
	Complete	0.25	0.00048		
	Total		0.0039249		
Pre-Code	Slight	11.16	0.0004464	266,048	233,602
	Moderate	12.96	0.0024624		
	Extensive	6.21	0.0059616		
	Complete	2.5	0.0048		
	Total		0.0136704		

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Table 8. Steel Building Damage Cost from an Earthquake with 100-Year Return Period

Design code	Damage state	Probability of damage (precentage)	Damage ratio (precentage)	Damage cost ($)	Repair energy (MJ)
High	Slight	23.34	0.0009336	83,640	73,440
	Moderate	11.39	0.0021641		
	Extensive	1.19	0.0011424		
	Complete	0.03	0.0000576		
	Total		0.0042977		
Moderate	Slight	21.21	0.0008484	169,110	148,486
	Moderate	18.38	0.0034922		
	Extensive	3.97	0.0038112		
	Complete	0.28	0.0005376		
	Total		0.0086894		
Pre-Code	Slight	16.01	0.0006404	735,719	645,995
	Moderate	33.76	0.0064144		
	Extensive	20.45	0.019632		
	Complete	5.79	0.0111168		
	Total		0.0378036		

As is seen, the difference from high code to moderate code is substantial, both with respect to repair costs and environmental impacts. In one year of its life, a typical concrete building experiences $1,218 of damage and consumes 1,968 MJ of energy.

Results and Discussion

A review of the results obtained for the concrete and steel buildings' annual damage and repair costs and associated environmental impacts showed that the population of concrete buildings had lower annual loss considering both high and moderate code standards. *HAZUS-MH* methodologies consider each building's location and underlying soil type. The fact that the results depict lower concrete building damage does not necessarily imply that concrete buildings behave better under seismic events. Additionally the seismic design codes for each building type progress at different rates, which could have resulted in the concrete building population exhibiting less estimated damage. The Northridge event was examined to determine the differences among each building's behavior. The results are somewhat subjective given the *HAZUS-MH* methodology and the assumptions made about each building type. The future research section of this paper discusses these issues and suggests analyses that may be used to better understand building environmental impacts when subjected to seismic events.

Table 9. Steel Building Damage Data from Annualized Seismic Damage

Design code	Damage state	Probability of damage (precentage)	Damage ratio (precentage)	Damage cost ($)	Repair energy (MJ)
High	Slight	1.53	0.0000612	1,894	1,663
	Moderate	0.19	0.0000361		
	Extensive	0	0		
	Complete	0	0		
	Total		0.0000973		
Moderate	Slight	1.15	0.000046	4,673	4,103
	Moderate	0.87	0.0001653		
	Extensive	0.03	0.0000288		
	Complete	0	0		
	Total		0.0002401		

Table 10. *HAZUS-MH* and Data for Damage Cost Calculations

Damage relationship	Value
Concrete—energy per replacement cost of building (MJ/$)	1.62
Concrete building total cost replacement value ($/SF)	183.87
Energy for building, 60-year lifespan (MJ/SF)	297

Table 11. Concrete Building Damage Cost from Northridge Earthquake

Design code	Damage state	Probability of damage (precentage)	Damage ratio (precentage)	Damage cost ($)	Repair energy (MJ)
High	Slight	9.65	0.000386	22,276	35,982
	Moderate	2.62	0.0004978		
	Extensive	0.42	0.0004032		
	Complete	0.06	0.0001152		
	Total		0.0014022		
Moderate	Slight	9.23	0.0003692	48,436	78,237
	Moderate	5.11	0.0009709		
	Extensive	1.48	0.0014208		
	Complete	0.15	0.000288		
	Total		0.0030489		
Pre-Code	Slight	12.56	0.0005024	167,727	270,924
	Moderate	11.29	0.0021451		
	Extensive	5.12	0.0049152		
	Complete	1.56	0.0029952		
	Total		0.0105579		

Building Energy Consumption when Subjected to Seismic Events

Table 14 illustrates the total environmental impacts when comparing steel to concrete. As is evident, the steel building outperforms the concrete building when considering each building's life-cycle assessment, although the *HAZUS-MH* results show that the annual loss from damage was greater for the steel population than the concrete building population. Cost, seismic resiliency, and environmental impacts all need to be considered to create a sustainable design. One needs to consider a higher initial investment and weighing its long-term benefits.

Table 14 represents the total LCA impact of each building, including their estimated annual seismic damage for 60 years.

Table 12. Concrete Building Damage Cost from a 100-Year Return Period Earthquake

Design code	Damage state	Probability of damage (precentage)	Damage ratio (precentage)	Damage cost ($)	Repair energy (MJ)
High	Slight	24.23	0.0009692	56,819	91,778
	Moderate	6.7	0.001273		
	Extensive	1.15	0.001104		
	Complete	0.12	0.0002304		
	Total		0.0035766		
Moderate	Slight	22.06	0.0008824	104,138	168,212
	Moderate	12.88	0.0024472		
	Extensive	3	0.00288		
	Complete	0.18	0.0003456		
	Total		0.0065552		
Pre-Code	Slight	20.15	0.000806	487,038	786,699
	Moderate	31.96	0.0060724		
	Extensive	18.15	0.017424		
	Complete	3.31	0.0063552		
	Total		0.0306576		

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Table 13. Concrete Building Annualized Seismic Damage Cost

Design code	Damage state	Probability of damage (precentage)	Damage ratio (precentage)	Damage cost ($)	Repair energy (MJ)
High	Slight	0.59	0.0000236	375	606
	Moderate	0	0		
	Extensive	0	0		
	Complete	0	0		
	Total		0.0000236		
Moderate	Slight	0.54	0.0000216	1,218	1,968
	Moderate	0.29	0.0000551		
	Extensive	0	0		
	Complete	0	0		
	Total		0.0000767		

Table 14. Overall Life-Cycle Impacts Including Seismic Damage

Building type	Design level	LCA energy MJ/SF	Annual earthquake repair energy (MJ/60 years)/SF	Overall environmental impact MJ/SF
Steel building	High	8,949	1.15	8,950.15
	Moderate	8,949	2.85	8,951.85
Concrete building	High	9,048	0.42	9,048.42
	Moderate	9,048	1.37	9,049.37

The results illustrate that the environmental effect from the seismic repair damage are a small portion of a building's total overall LCA impact. The steel building repair damage is approximately 0.58% of its total embodied energy and the concrete building's repair damage is approximately 0.14% of its total embodied energy (considering the high design level). Whereas this repair energy is a small portion of the building's total LCA, future research needs to be completed to effectively understand the environmental implications of seismic events. The comparison made in this study between the two building types showed that they are similar. Additional research should examine nonstructural damage and different building uses. The purpose of this study was to provide a method for use with all buildings types to examine the effect of the repair damage required after earthquake events.

Value of Earthquake Mitigation Efforts

The data presented in Table 15 demonstrates how earthquake mitigation efforts save money and reduce energy consumption considering the damage consequences. The design code affects damage and repair costs; upgrading to a more current code level decreases damage costs and reduces energy consumption. This study examined the differences between steel- and concrete-framed structures. Tables 15, 16, and 17 present cost and energy consumption differences and the savings that could result from choosing between the two building types.

Table 15 displays information previously discussed in this paper as a reference for Tables 16 and 17. As shown, the concrete frame building examined in this study has a higher annualized damage cost and energy consumption than the steel framed building for projected earthquake damage.

Table 15. Annualized Earthquake Event Comparison

Building type	Design level	Damage cost ($)	Energy consumption from repair (MJ)
Concrete	High	1,894	1,663
	Moderate	4,673	4,103
Steel	High	375	606
	Moderate	1,218	1,968

Table 16. Annualized Savings: Upgrading to a Higher Code Design Level

Building type	Money saved from upgrading to high design level ($)	Energy consumption saved from upgrading to high design level (MJ)
Steel	843	1,362
Concrete	2,779	2,440

Table 16 provides the savings that would result if a building was designed to a higher design code level. Upgrading from a moderate design code to a high code design level creates savings of approximately $843 and reduces annual energy consumption by 1,362 MJ for a steel-frame office building. The same information is shown for a concrete-frame building. Multiplying these savings by the number of office buildings in California shows a significant reduction in the environmental impact from the repair damage savings. The population of moderately designed steel buildings that *HAZUS-MH* provides for the Los Angeles area alone is 28,276. Multiplying this number by the annual energy consumption for repair damage results in 116,016,428 MJ of energy. The savings subsequently provided from converting to a high design code level multiplied by this population of moderately designed buildings in the Los Angeles area gives a savings of 38,511,912 MJ of energy. This same examination may be done using dollar amounts. Although upgrading a building to a higher code standard has the potential to reduce the energy consumption and help reduce repair costs attributable to future earthquake events, the costs to upgrade a building to a higher code level must also be factored into the decision on whether or not to upgrade.

Upgrading a building to a higher code standard greatly reduces energy consumption and helps reduce repair costs, although the cost to achieve a higher code standard building needs to be considered. The dollar amounts and energy consumed discussed only consider damage that may happen. Costs to upgrade a building to a higher code level must be weighed with the damage repair consequences.

This study only examined a single office building and found that the results were minimal when comparing a building's overall LCA to the environmental results of the earthquake damage repair. However, this observation considers a single building. Sustainability should consider the entire population of buildings that may be affected, which may add up to a substantial amount of energy use and earthquake damage repair costs.

Future Research Recommendations

This study examined two building frames and did not include nonstructural building items. Future research should examine full building envelope LCAs to combine them with estimated seismic damage repair. The structural frame has the most substantial effect

Table 17. Annualized Savings: Difference between Steel and Concrete Building Frame

Building type and code design level	Money saved from changing building types ($)	Energy consumption saved from changing building types (MJ)
Concrete moderate to steel moderate	3,455	2,135
Concrete high to steel high	1,519	1,057

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when comparing the two structure's seismic resiliency, although the nonstructural components of a building would additionally contribute to a building's LCA effect.

Future research should examine the environmental impacts of nonstructural damage to a building's operating costs. This paper determined that a building's operational energy is the largest contributor to its overall LCA, and seismic damage to any system that may affect a building's energy use may result in higher costs and greater environmental impacts. *HAZUS-MH* may be used to estimate nonstructural damage, and this information may be used to examine its role in affecting a building's operational energy usage.

Future research should examine the cost and energy consumption required to upgrade a building to a higher code design level, and the cost savings resulting from an earthquake event. The current research examined only cost savings associated with damage repair and did not consider the amount of energy and the costs associated with upgrading a building to a higher design level.

The writers also suggest using *HAZUS-MH* for individual seismic evaluation. This study used the damage probability for each building type as represented by general building population values. Although such values are reasonable average estimations of damage, the location and the epicenter of earthquakes affect the estimated damage results. Examination of an individual building would eliminate these issues and provide a more accurate damage and repair estimation.

LCAs are independently challenging without considering natural hazard damage effects; programs such as *Athena* should incorporate natural hazard damages to ease the collection of input data and the cost estimates for damage repair.

Additionally, more incentives should be awarded to buildings that include natural hazard mitigation design strategies. Environmental programs such as LEED should consider including the effects caused by natural hazards in their standardizations. Although this research determined that structural repair costs were low when compared with buildings' overall costs, this examination could go further by including nonstructural components that may reduce costs and future environmental impacts. Possible alternatives to reducing the effects of a building's usage phase could examine decreasing heating, lighting, and electricity usages.

Summary and Conclusions

This study proposed a method for combining seismic analysis and LCA results to obtain a building's overall environmental impact, including the costs of repairing the damage caused by earthquakes. Using *Athena* and *HAZUS-MH*, this study explored the differences between steel and concrete buildings using the proposed method. The code design level was determined to greatly affect a building's repair and damage estimations. This study examined two case study buildings and found specific results on the basis of several assumptions. Using these assumptions, this study determined that the difference between steel and concrete office buildings was minimal when comparing their overall LCA impacts, similar to the finding by Guggemos and Horvath (2005). Using the *HAZUS-MH* software tool, the general population of concrete buildings was determined to have a lower damage cost compared with the general population of steel buildings. This study also determined that the operational energy contributed approximately 90% of the total energy use compared with a building's overall LCA. Examining the cost and environmental impacts that a building exhibits by using a cradle-to-grave analysis and seismic damage assessment will help reduce material consumption and construction activities before and after an earthquake event occurs.

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